Worksheet 4 and 5

Problem 4.3

```
library(mnormt)
rmnorm(n=10,varcov=diag(10))
```

```
##               [,1]        [,2]       [,3]        [,4]        [,5]        [,6]
##  [1,] -0.530339489 -0.66161153  0.1269725  0.21014753 -0.51241099  0.52904651
##  [2,]  0.705134502  0.11679785  0.3264102 -0.03449778 -0.06978828 -0.68273340
##  [3,] -0.433022630  1.08898506  0.9003740 -1.24368741  0.54294049 -1.17611277
##  [4,] -1.250282317  0.46452270  1.1821021  1.00667266 -0.85721666 -0.45091775
##  [5,]  1.023265647  0.43645462  1.6074890  0.36833866 -0.59093632 -0.01367928
##  [6,] -0.432554466 -0.57760543 -0.3746249  2.24620264  0.18888890 -0.01966443
##  [7,]  2.350634751  0.06385632  0.3493676 -0.90001858  1.00361542 -0.28130773
##  [8,]  0.430674669  0.47140267  1.8083064  0.34475322 -0.17913662 -0.28799328
##  [9,] -0.246710394 -0.25007278  0.6073946  0.04287341  0.29839865 -0.45395287
## [10,]  0.001326652 -1.75273972 -0.3827975 -0.90695340  0.58730867  1.81632347
##              [,7]        [,8]        [,9]      [,10]
##  [1,] -1.26724456 -2.08011053 -1.45621083 -1.1039785
##  [2,] -0.93026089  1.26010158 -0.32856494  1.1461805
##  [3,] -0.08701058 -0.10202044  0.49114495 -0.4751761
##  [4,]  0.57254071 -0.13548735 -0.79524070 -2.1284742
##  [5,]  0.01308251  0.40483804 -0.07659356 -0.4621673
##  [6,]  0.62043832  2.01915901 -0.60609880  0.5673305
##  [7,] -0.18470835  0.77935273 -0.31688107 -0.5957618
##  [8,]  0.63800989 -0.06788795 -0.74469950  0.8985744
##  [9,] -0.64979396  1.61856429 -1.26405625 -1.6159468
## [10,] -0.54193979  0.42472904 -1.15872759  1.7175877
```

```
myrmnorm=function(n,mu,Sigma){
    d<-as.numeric(nrow(mu))
    average<-matrix(rep(0,times=n*d),nrow=d)
    for (i in 1:d){
        temp<-rnorm(n,mean=0,sd=1)
        average[i,]<-temp
    }
    vect<-eigen(Sigma)$vectors%*%diag(sqrt(eigen(Sigma)$values))
    tem<-matrix(rep(mu,times=n),ncol=n)
    print(tem)
    print(tem+vect%*%average)
}
mean1<-matrix(c(1,2,3),ncol=1)
vari<-matrix(c(1,0,0,0,1,0,0,0,1),nrow=3)
myrmnorm(10,mean1,vari)
```

```
##      [,1] [,2] [,3] [,4] [,5] [,6] [,7] [,8] [,9] [,10]
## [1,]    1    1    1    1    1    1    1    1    1     1
## [2,]    2    2    2    2    2    2    2    2    2     2
```

```
## [3,]    3    3    3    3    3    3    3    3    3     3
##            [,1]      [,2]     [,3]      [,4]     [,5]       [,6]     [,7]
## [1,] -0.1319722 1.6540265 2.032954 -1.853935 1.042465 -0.5130822 1.005725
## [2,]  2.5775187 0.9011049 2.667948  3.007337 1.384427  3.3214038 3.826914
## [3,]  1.9873260 2.8726665 2.564695  3.645980 2.208604  2.2517154 2.589623
##          [,8]      [,9]     [,10]
## [1,] 1.319978 0.3222031 0.2764732
## [2,] 4.933113 0.6847112 2.0093948
## [3,] 4.084202 2.5565390 3.6030428
```

Problem 5.2

```
library(mnormt)
mu<-matrix(c(0,0),nrow=2)
sigm<-matrix(c(2,2,2,4),nrow=2)
data<-rmnorm(500,mean=mu,varcov=sigm)
head(data)
```

```
##            [,1]       [,2]
## [1,]  1.5279932  2.8073446
## [2,]  1.0944651  0.1749712
## [3,] -0.3305746  1.3391141
## [4,] -1.1966477 -0.8244819
## [5,]  1.0231356  2.5030447
## [6,]  0.5739026 -0.7703792
```

```
library(whitening)
```

```
## Loading required package: corpcor
```

```
zca<- whiten(data,method="ZCA")
zcacor<- whiten(data,method="ZCA-cor")
pca<- whiten(data,method="PCA")
pcacor<- whiten(data,method="PCA-cor")
chol<-whiten(data,method="Cholesky")
head(zca);head(zcacor);head(pca);head(pcacor);head(chol)
```

```
##              L1         L2
## [1,]  0.6189983  1.2477583
## [2,]  0.9863872 -0.2177855
## [3,] -0.7096468  0.9085457
## [4,] -0.8911982 -0.1452377
## [5,]  0.2301285  1.2128469
## [6,]  0.7722313 -0.6361764
```

```
##              L1          L2
## [1,]  0.7120435  1.19710219
## [2,]  0.9669812 -0.29212577
## [3,] -0.6385402  0.95985430
## [4,] -0.8996591 -0.07708224
## [5,]  0.3216451  1.19184775
## [6,]  0.7216451 -0.69302953
```

```
##              L1          L2
## [1,]  1.3901749  0.08644898
## [2,]  0.3036421 -0.96342704
```

```
## [3,]  0.4330472  1.06842127
## [4,] -0.5706934  0.69974081
## [5,]  1.1658187  0.40598437
## [6,] -0.1657960 -0.98669818
```

```
##               L1         L2
## [1,]  1.34996988  0.3429883
## [2,]  0.47719488 -0.8903231
## [3,]  0.22720337  1.1302356
## [4,] -0.69066042  0.5816497
## [5,]  1.07020105  0.6153262
## [6,]  0.02023427 -1.0003260
```

```
##              L1         L2
## [1,]  1.1038805  0.8494159
## [2,]  0.7906833 -0.6286574
## [3,] -0.2388197  1.1278382
## [4,] -0.8645039  0.2606936
## [5,]  0.7391522  0.9887420
## [6,]  0.4146092 -0.9105827
```

```
cor(zca);cor(zcacor);cor(pca);cor(pcacor);cor(chol)
```

```
##               L1            L2
## L1  1.000000e+00 -2.693876e-16
## L2 -2.693876e-16  1.000000e+00
```

```
##              L1           L2
## L1 1.000000e+00 1.271492e-16
## L2 1.271492e-16 1.000000e+00
```

```
##              L1           L2
## L1 1.000000e+00 1.027943e-15
## L2 1.027943e-15 1.000000e+00
```

```
##              L1           L2
## L1 1.000000e+00 1.137624e-15
## L2 1.137624e-15 1.000000e+00
```

```
##              L1           L2
## L1 1.000000e+00 8.073772e-16
## L2 8.073772e-16 1.000000e+00
```

```
cov(zca);cov(zcacor);cov(pca);cov(pcacor);cov(chol)
```

```
##               L1            L2
## L1  1.000000e+00 -2.693876e-16
## L2 -2.693876e-16  1.000000e+00
```

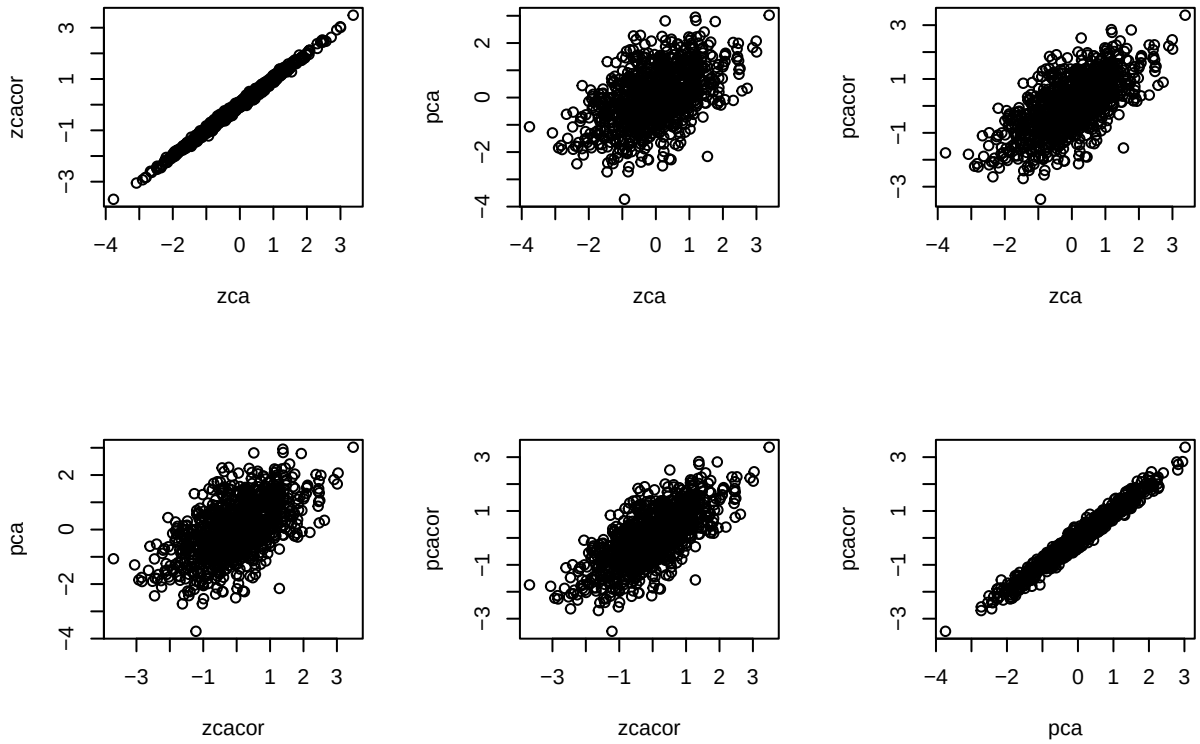
```
##              L1           L2
## L1 1.000000e+00 1.271492e-16
## L2 1.271492e-16 1.000000e+00
```

```
##              L1           L2
## L1 1.000000e+00 1.027943e-15
## L2 1.027943e-15 1.000000e+00
```

```
##              L1           L2
## L1 1.000000e+00 1.137624e-15
## L2 1.137624e-15 1.000000e+00
```

```
##               L1           L2
## L1 1.000000e+00 8.073772e-16
## L2 8.073772e-16 1.000000e+00
```

```
par(mfrow=c(2,3))
plot(zca,zcacor);plot(zca,pca);plot(zca,pcacor)
plot(zcacor,pca);plot(zcacor,pcacor);plot(pca,pcacor)
```




```
par(mfrow=c(3,2))
plot(data,zca);plot(data,zcacor)
plot(data,pca);plot(data,pcacor);plot(data,chol)

cor(data,zca);cor(data,zcacor)
```

```
##              L1        L2
## [1,] 0.8985115 0.4389499
## [2,] 0.2976912 0.9546622
```

```
##              L1        L2
## [1,] 0.9292749 0.3693890
## [2,] 0.3693890 0.9292749
```

```
cor(data,pca);cor(data,pcacor);cor(data,chol)
```

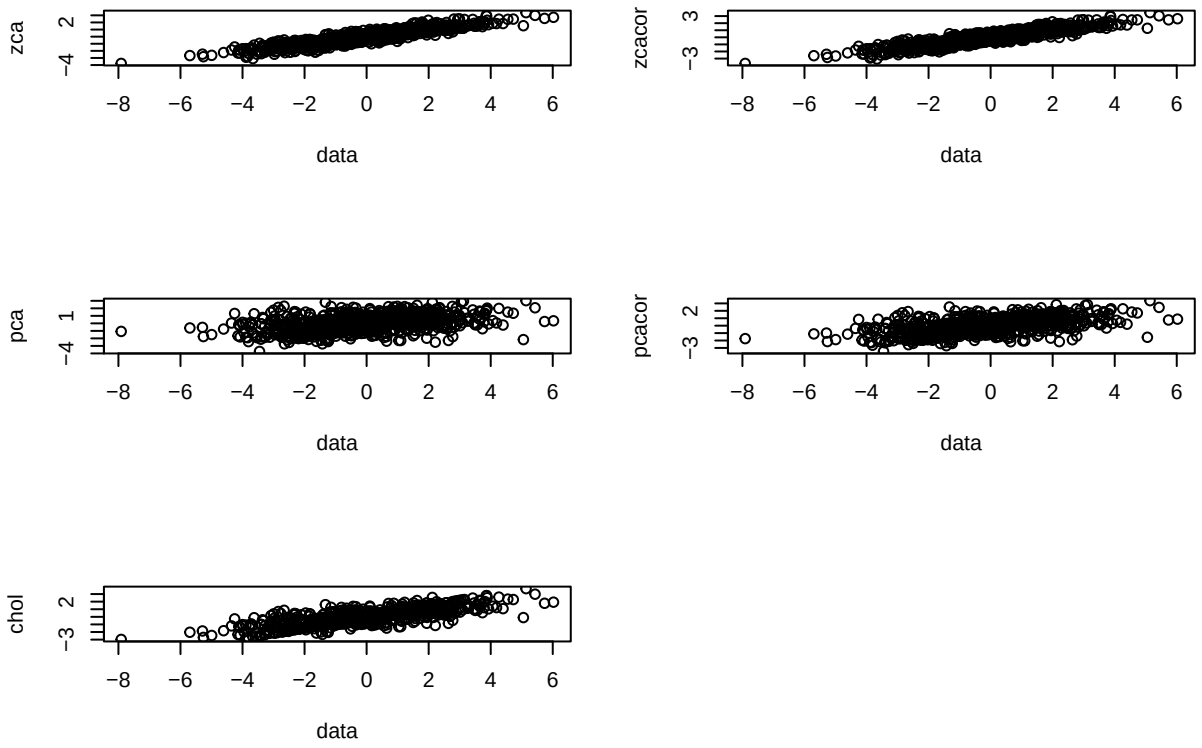
```
##              L1         L2
## [1,] 0.8288499 -0.5594711
## [2,] 0.9758219  0.2185671
```

```
##             L1          L2
```

```
## [1,] 0.918294 -0.3958991
## [2,] 0.918294  0.3958991
```

```
##             L1           L2
## [1,] 1.0000000 8.696768e-16
## [2,] 0.6865278 7.271035e-01
```

```
par(mfrow=c(3,2))
```




```
image(cor(data,zca));image(cor(data,zcacor))
image(cor(data,pca));image(cor(data,pcacor));image(cor(data,chol))

temp4<-cor(data,zca)
temp4[1,1]+temp4[2,2]
```

```
##       L1
## 1.853174
```

```
temp4<-cor(data,zcacor)
temp4[1,1]+temp4[2,2]
```

```
##      L1
## 1.85855
```

```
temp4<-cor(data,pca)
temp4[1,1]+temp4[2,2]
```

```
##       L1
## 1.047417
```

```
temp4<-cor(data,pcacor)
temp4[1,1]+temp4[2,2]
```

```
##       L1
## 1.314193
```

```
#Since they are same dimensional, we dont need to divide d.

temp4<-t(cor(data,zca))%*%cor(data,zca)
temp4[1,1]^2+temp4[2,2]^2
```

```
## [1] 2.021656
```

```
temp4<-t(cor(data,zcacor))%*%cor(data,zcacor)
temp4[1,1]^2+temp4[2,2]^2
```

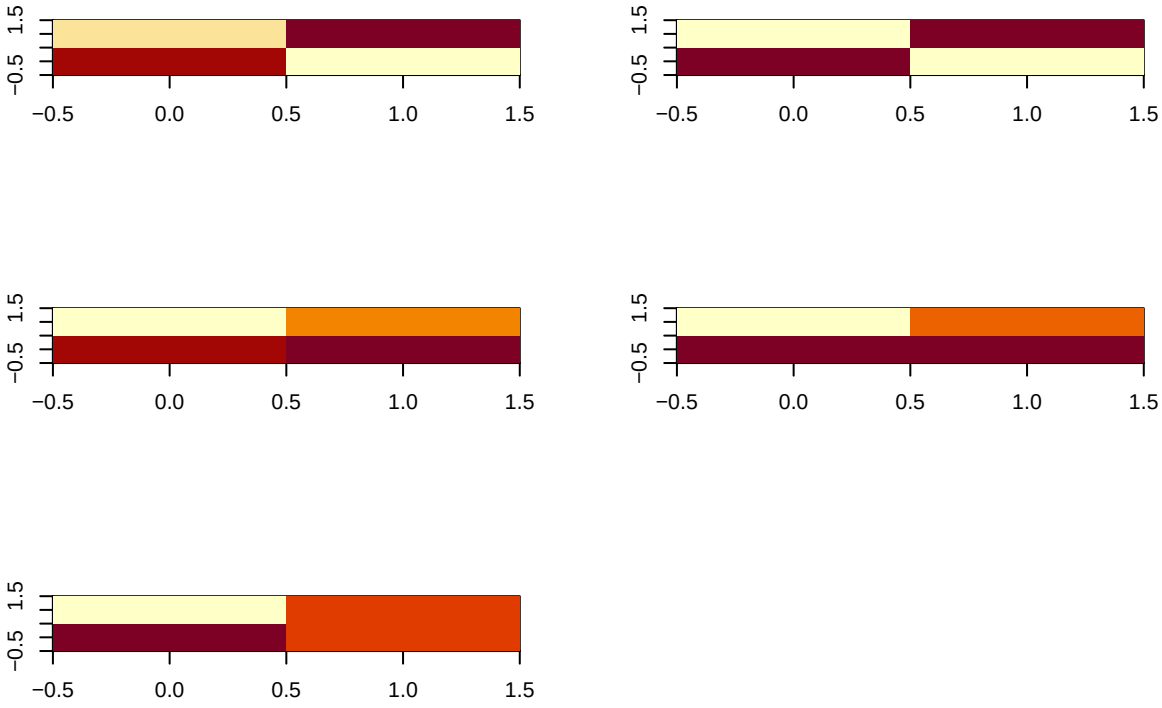
```
## [1] 2
```

```
temp4<-t(cor(data,pca))%*%cor(data,pca)
temp4[1,1]^2+temp4[2,2]^2
```

```
## [1] 2.817206
```

```
temp4<-t(cor(data,pcacor))%*%cor(data,pcacor)
temp4[1,1]^2+temp4[2,2]^2
```

```
## [1] 2.942641
```



Problem 5.3

```r
data(iris)
summary(iris)
```

```
##   Sepal.Length    Sepal.Width     Petal.Length    Petal.Width
##  Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100
##  1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300
##  Median :5.800   Median :3.000   Median :4.350   Median :1.300
##  Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199
##  3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800
##  Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500
##        Species
##  setosa    :50
##  versicolor:50
##  virginica :50
##
##
##
```

```r
head(iris)
```

```
##   Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1          5.1         3.5          1.4         0.2  setosa
## 2          4.9         3.0          1.4         0.2  setosa
## 3          4.7         3.2          1.3         0.2  setosa
## 4          4.6         3.1          1.5         0.2  setosa
## 5          5.0         3.6          1.4         0.2  setosa
## 6          5.4         3.9          1.7         0.4  setosa
```

```r
data<-matrix(c(iris[,1],iris[,2],iris[,3],iris[,4]),nrow=length(iris[,1]))
head(data)
```

```
##      [,1] [,2] [,3] [,4]
## [1,]  5.1  3.5  1.4  0.2
## [2,]  4.9  3.0  1.4  0.2
## [3,]  4.7  3.2  1.3  0.2
## [4,]  4.6  3.1  1.5  0.2
## [5,]  5.0  3.6  1.4  0.2
## [6,]  5.4  3.9  1.7  0.4
```

```r
variable<-iris[,5]
head(variable)
```

```
## [1] setosa setosa setosa setosa setosa setosa
## Levels: setosa versicolor virginica
```

```r
pr<-prcomp(data)
print(pr);print(pr$x) #None of them capture 95% of the variation.
```

```
## Standard deviations (1, .., p=4):
## [1] 2.0562689 0.4926162 0.2796596 0.1543862
##
## Rotation (n x k) = (4 x 4):
##              PC1         PC2         PC3        PC4
## [1,]  0.36138659 -0.65658877  0.58202985  0.3154872
## [2,] -0.08452251 -0.73016143 -0.59791083 -0.3197231
## [3,]  0.85667061  0.17337266 -0.07623608 -0.4798390
## [4,]  0.35828920  0.07548102 -0.54583143  0.7536574
```

```
##                PC1          PC2          PC3           PC4
##   [1,] -2.684125626 -0.319397247  0.027914828  0.0022624371
##   [2,] -2.714141687  0.177001225  0.210464272  0.0990265503
##   [3,] -2.888990569  0.144949426 -0.017900256  0.0199683897
##   [4,] -2.745342856  0.318298979 -0.031559374 -0.0755758166
##   [5,] -2.728716537 -0.326754513 -0.090079241 -0.0612585926
##   [6,] -2.280859633 -0.741330449 -0.168677658 -0.0242008576
##   [7,] -2.820537751  0.089461385 -0.257892158 -0.0481431065
##   [8,] -2.626144973 -0.163384960  0.021879318 -0.0452978706
##   [9,] -2.886382732  0.578311754 -0.020759570 -0.0267447358
##  [10,] -2.672755798  0.113774246  0.197632725 -0.0562954013
##  [11,] -2.506947091 -0.645068899  0.075318009 -0.0150199245
##  [12,] -2.612755231 -0.014729939 -0.102150260 -0.1563792078
##  [13,] -2.786109266  0.235112000  0.206844430 -0.0078879115
##  [14,] -3.223803744  0.511394587 -0.061299672 -0.0216798118
##  [15,] -2.644750390 -1.178764636  0.151627524  0.1592097177
##  [16,] -2.386039034 -1.338062330 -0.277776903  0.0065515459
##  [17,] -2.623527875 -0.810679514 -0.138183228  0.1677347372
##  [18,] -2.648296706 -0.311849145 -0.026668316  0.0776281796
##  [19,] -2.199820324 -0.872839039  0.120305523  0.0270518681
##  [20,] -2.587986400 -0.513560309 -0.213665172 -0.0662726502
##  [21,] -2.310256215 -0.391345936  0.239444043 -0.0150707908
##  [22,] -2.543705229 -0.432996063 -0.208457232  0.0410654027
##  [23,] -3.215939416 -0.133468070 -0.292396751  0.0044821251
##  [24,] -2.302733182 -0.098708855 -0.039123259  0.1483525893
##  [25,] -2.355754049  0.037281860 -0.125021083 -0.3003309039
##  [26,] -2.506668907  0.146016880  0.253420042  0.0346074722
##  [27,] -2.468820073 -0.130951489 -0.094910576  0.0574497158
##  [28,] -2.562319906 -0.367718857  0.078494205 -0.0141727423
##  [29,] -2.639534715 -0.312039980  0.145908896  0.0657834667
##  [30,] -2.631989387  0.196961225 -0.040771079 -0.1239833064
##  [31,] -2.587398477  0.204318491  0.077222989 -0.0604622767
##  [32,] -2.409932497 -0.410924264  0.145524972  0.2316284917
##  [33,] -2.648862334 -0.813363820 -0.225669150 -0.2813723471
##  [34,] -2.598736749 -1.093145759 -0.157810813 -0.0953488583
##  [35,] -2.636926878  0.121322348  0.143049582  0.0190703413
##  [36,] -2.866241652 -0.069364472  0.164332307  0.1625984463
##  [37,] -2.625238050 -0.599370021  0.268350376  0.1764412129
##  [38,] -2.800684115 -0.268643738 -0.093699082 -0.1681730544
##  [39,] -2.980502044  0.487958344 -0.072927046 -0.0107331474
##  [40,] -2.590006314 -0.229043837  0.080082303 -0.0137491513
##  [41,] -2.770102426 -0.263527534 -0.077247693  0.0940633590
##  [42,] -2.849368705  0.940960574  0.349230377  0.3199874870
##  [43,] -2.997406547  0.341926057 -0.192509212 -0.0746777682
##  [44,] -2.405614485 -0.188871429 -0.263867946  0.1762088905
##  [45,] -2.209489238 -0.436663142 -0.298742746 -0.1828425025
##  [46,] -2.714451427  0.250208204  0.097678144  0.1428435736
##  [47,] -2.538148259 -0.503771144 -0.166705637 -0.1896222914
##  [48,] -2.839462168  0.227945569 -0.083726849 -0.0595642283
##  [49,] -2.543085750 -0.579410022  0.017115024 -0.0465686438
##  [50,] -2.703359782 -0.107706082  0.089294008  0.0346583385
##  [51,]  1.284825689 -0.685160470  0.406568025  0.0185252879
##  [52,]  0.932488532 -0.318333638  0.018014187  0.0005665121
##  [53,]  1.464302322 -0.504262815  0.338325765 -0.0016531759
```

```
##  [54,]  0.183317720  0.827959012  0.179591392  0.0935668402
##  [55,]  1.088103258 -0.074590675  0.307757896  0.1120205742
##  [56,]  0.641669084  0.418246872 -0.041076091 -0.2431167665
##  [57,]  1.095060663 -0.283468270 -0.169810240 -0.0835565724
##  [58,] -0.749122670  1.004890961 -0.012302919 -0.0179077226
##  [59,]  1.044131826 -0.228361900  0.415336085 -0.0391345020
##  [60,] -0.008745404  0.723081905 -0.281141431 -0.0056189179
##  [61,] -0.507840884  1.265971191  0.269817183  0.0455624408
##  [62,]  0.511698557  0.103981235 -0.130547750  0.0507192325
##  [63,]  0.264976508  0.550036464  0.694146830  0.0571855195
##  [64,]  0.984934510  0.124817854  0.062114408 -0.1694962546
##  [65,] -0.173925372  0.254854209 -0.090457691  0.1252172921
##  [66,]  0.927860781 -0.467179494  0.314620976  0.0998031365
##  [67,]  0.660283762  0.352969666 -0.328027528 -0.1878786215
##  [68,]  0.236104993  0.333610767  0.271161837 -0.2137573696
##  [69,]  0.944733728  0.543145551  0.499519046  0.2571921772
##  [70,]  0.045226976  0.583834377  0.235002105 -0.0415766476
##  [71,]  1.116283177  0.084616852 -0.459620991 -0.0750315529
##  [72,]  0.357888418  0.068925032  0.229853888  0.1229976041
##  [73,]  1.298183875  0.327787308  0.347854352  0.0008883706
##  [74,]  0.921728922  0.182737794  0.231071778 -0.2882554293
##  [75,]  0.714853326 -0.149055944  0.321800937  0.0417197556
##  [76,]  0.900174373 -0.328504474  0.316209074  0.1002267276
##  [77,]  1.332024437 -0.244440876  0.521702780  0.0353331921
##  [78,]  1.557802155 -0.267495447  0.164920984  0.0699692823
##  [79,]  0.813290650  0.163350301 -0.035424505 -0.0297114340
##  [80,] -0.305583778  0.368262190  0.318491581  0.0745696136
##  [81,] -0.068126492  0.705172132  0.244213810  0.0068308422
##  [82,] -0.189622472  0.680286764  0.306420561 -0.0205510016
##  [83,]  0.136428712  0.314032438  0.177242766  0.0329419128
##  [84,]  1.380026436  0.420954287 -0.016167128 -0.1783044629
##  [85,]  0.588006443  0.484287420 -0.444433499 -0.2509760601
##  [86,]  0.806858313 -0.194182315 -0.388963063 -0.1142072433
##  [87,]  1.220690882 -0.407619594  0.237167010  0.0312171829
##  [88,]  0.815095236  0.372037060  0.614720843  0.1540209998
##  [89,]  0.245957680  0.268524397 -0.188366812 -0.1466745117
##  [90,]  0.166413217  0.681926725  0.060009226  0.0296222195
##  [91,]  0.464800288  0.670711545  0.024306856 -0.2696514282
##  [92,]  0.890815198  0.034464444  0.009946933 -0.1534846663
##  [93,]  0.230548024  0.404385848  0.229410241  0.0169303245
##  [94,] -0.704531759  1.012248228  0.105691149  0.0456133071
##  [95,]  0.356981495  0.504910093 -0.016617170 -0.0987414793
##  [96,]  0.331934480  0.212654684 -0.083204291 -0.2384754337
##  [97,]  0.376215651  0.293218929 -0.077996351 -0.1311373808
##  [98,]  0.642576008 -0.017738190  0.205394967 -0.0213776830
##  [99,] -0.906469865  0.756093367  0.012599648  0.2325348443
## [100,]  0.299000842  0.348897806 -0.010581660 -0.0511811717
## [101,]  2.531192728  0.009849109 -0.760165427 -0.0290555728
## [102,]  1.415235877  0.574916348 -0.296322527 -0.0153046739
## [103,]  2.616676016 -0.343903151  0.110787883  0.0657720412
## [104,]  1.971531053  0.179727904 -0.108424662 -0.2367909342
## [105,]  2.350005920  0.040260947 -0.285389563 -0.0001706333
## [106,]  3.397038736 -0.550836673  0.348437556 -0.1123716532
## [107,]  0.521232244  1.192758727 -0.545659296 -0.0981266196
```

```
## [108,]  2.932587069 -0.355500003  0.420239936 -0.2571910322
## [109,]  2.321228817  0.243831502  0.348304395 -0.0786746130
## [110,]  2.916750967 -0.782791949 -0.423335418  0.1109820710
## [111,]  1.661774154 -0.242228408 -0.242440190  0.1210405518
## [112,]  1.803401953  0.215637617  0.037648168  0.0780198444
## [113,]  2.165591796 -0.216275585 -0.033326642  0.1630614782
## [114,]  1.346163579  0.776818347 -0.281902882  0.1404408688
## [115,]  1.585928224  0.539640714 -0.629029326  0.3295517284
## [116,]  1.904456375 -0.119250692 -0.479639820  0.2196212627
## [117,]  1.949689059 -0.041943260 -0.044186168 -0.1576819073
## [118,]  3.487055364 -1.175739330 -0.133894874 -0.3092195730
## [119,]  3.795645422 -0.257322973  0.513767764  0.0538460965
## [120,]  1.300791713  0.761149636  0.344995038 -0.0458247549
## [121,]  2.427817913 -0.378196013 -0.219119324  0.1854292644
## [122,]  1.199001105  0.606091528 -0.511855509  0.0609591171
## [123,]  3.499920039 -0.460674099  0.573182243 -0.1402279544
## [124,]  1.388766132  0.204399327  0.064522757  0.1630409774
## [125,]  2.275430504 -0.334990606 -0.286150091 -0.0603719696
## [126,]  2.614090474 -0.560901355  0.205534524 -0.2407049865
## [127,]  1.258508161  0.179704795 -0.045847704  0.1475038465
## [128,]  1.291132059  0.116668651 -0.231256463  0.0040266077
## [129,]  2.123608723  0.209729477 -0.154180024  0.0528273230
## [130,]  2.388003016 -0.464639805  0.449530192 -0.2315240534
## [131,]  2.841672778 -0.375269167  0.498898076 -0.0223364626
## [132,]  3.230673661 -1.374165087  0.114548205 -0.2529019234
## [133,]  2.159437642  0.217277579 -0.208763167  0.1281930656
## [134,]  1.444161242  0.143413410  0.153233888 -0.1909963579
## [135,]  1.781294810  0.499901681  0.172875189 -0.5054344118
## [136,]  3.076499932 -0.688085678  0.335592292  0.3098280446
## [137,]  2.144243314 -0.140064201 -0.734878937  0.0555419691
## [138,]  1.905098149 -0.049300526 -0.162180236 -0.2212029369
## [139,]  1.169326339  0.164990262 -0.281835840  0.0204617872
## [140,]  2.107611143 -0.372287872 -0.027291132  0.2106217858
## [141,]  2.314154705 -0.183651279 -0.322693747  0.2776537774
## [142,]  1.922267801 -0.409203467 -0.113586596  0.5053049669
## [143,]  1.415235877  0.574916348 -0.296322527 -0.0153046739
## [144,]  2.563013375 -0.277862603 -0.292569525  0.0579127477
## [145,]  2.418746183 -0.304798198 -0.504482664  0.2410910005
## [146,]  1.944109795 -0.187532303 -0.177825091  0.4261959400
## [147,]  1.527166615  0.375316983  0.121898172  0.2543674420
## [148,]  1.764345717 -0.078858855 -0.130481631  0.1370012739
## [149,]  1.900941614 -0.116627959 -0.723251563  0.0445953047
## [150,]  1.390188862  0.282660938 -0.362909648 -0.1550386282
```

```
cov(pr$x)
```

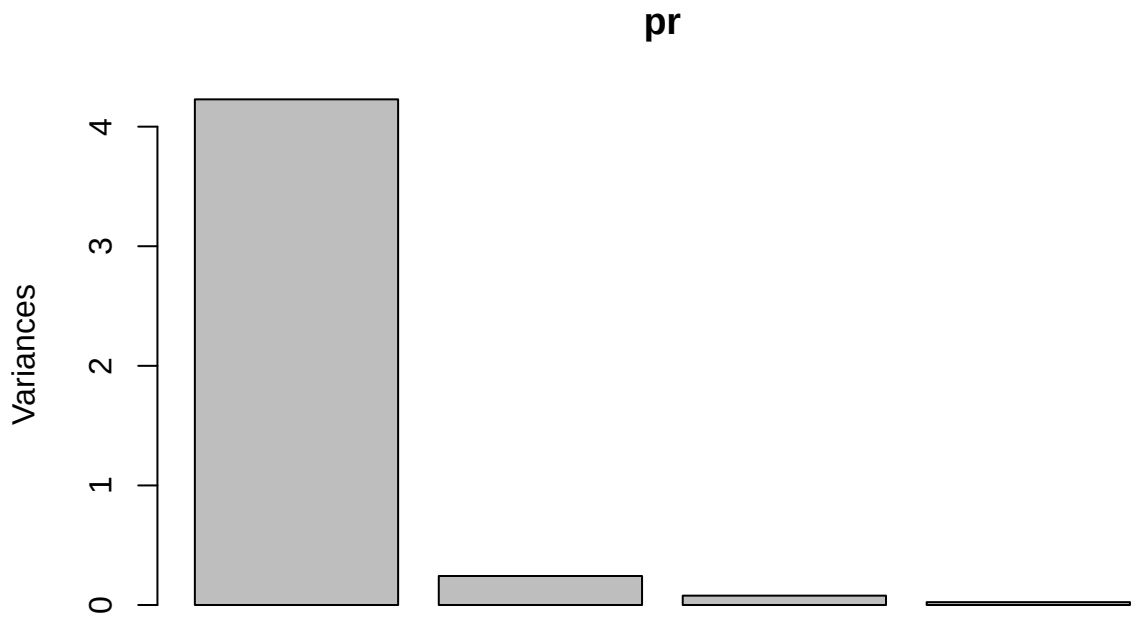
```
##                PC1           PC2           PC3           PC4
## PC1  4.228242e+00 -3.306941e-16  1.020681e-15  7.375519e-16
## PC2 -3.306941e-16  2.426707e-01 -3.362273e-17 -1.910476e-16
## PC3  1.020681e-15 -3.362273e-17  7.820950e-02  6.759155e-18
## PC4  7.375519e-16 -1.910476e-16  6.759155e-18  2.383509e-02
```

```
summary(pr)
```

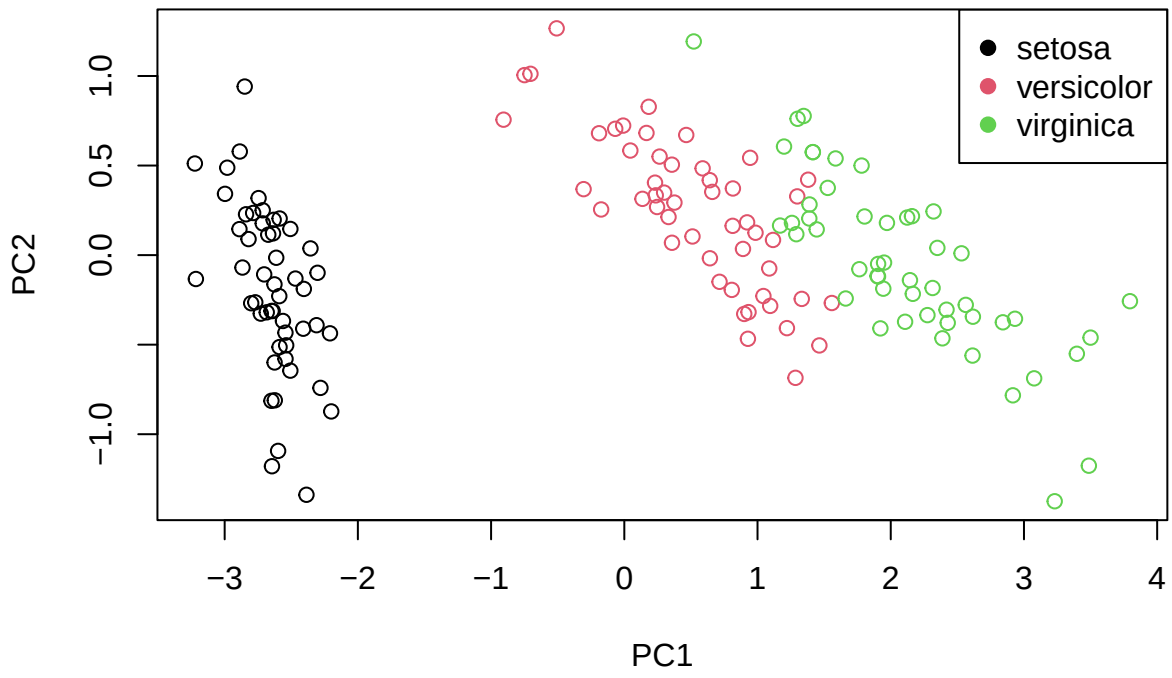
```
## Importance of components:
```

```
##                             PC1     PC2    PC3     PC4
## Standard deviation     2.0563 0.49262 0.2797 0.15439
## Proportion of Variance 0.9246 0.05307 0.0171 0.00521
## Cumulative Proportion  0.9246 0.97769 0.9948 1.00000
```

```
par(mfrow=c(1,1))
plot(pr)
```




```
plot(pr$x[,1],pr$x[,2],xlab="PC1",ylab="PC2",col=variable)
legend("topright",legend=levels(variable),col=c(1,2,3),pch=19)
```




```
sigma<-cov(data)
U<-eigen(sigma)$vectors
head(U)
```

```
##              [,1]        [,2]        [,3]       [,4]
## [1,]  0.36138659  0.65658877  0.58202985  0.3154872
## [2,] -0.08452251  0.73016143 -0.59791083 -0.3197231
## [3,]  0.85667061 -0.17337266 -0.07623608 -0.4798390
## [4,]  0.35828920 -0.07548102 -0.54583143  0.7536574
```

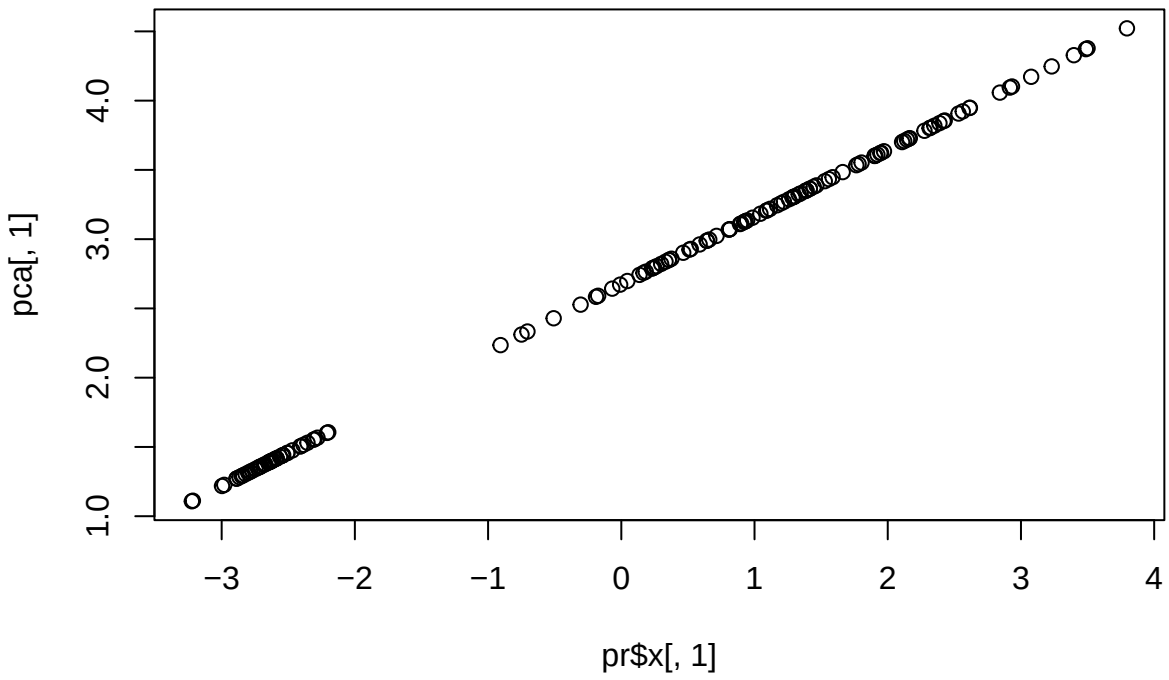
```
pca<-data%*%t(U)
eigen(sigma)$values/sum(diag(pca))
```

```
## [1] 0.426363677 0.024470217 0.007886420 0.002403462
```

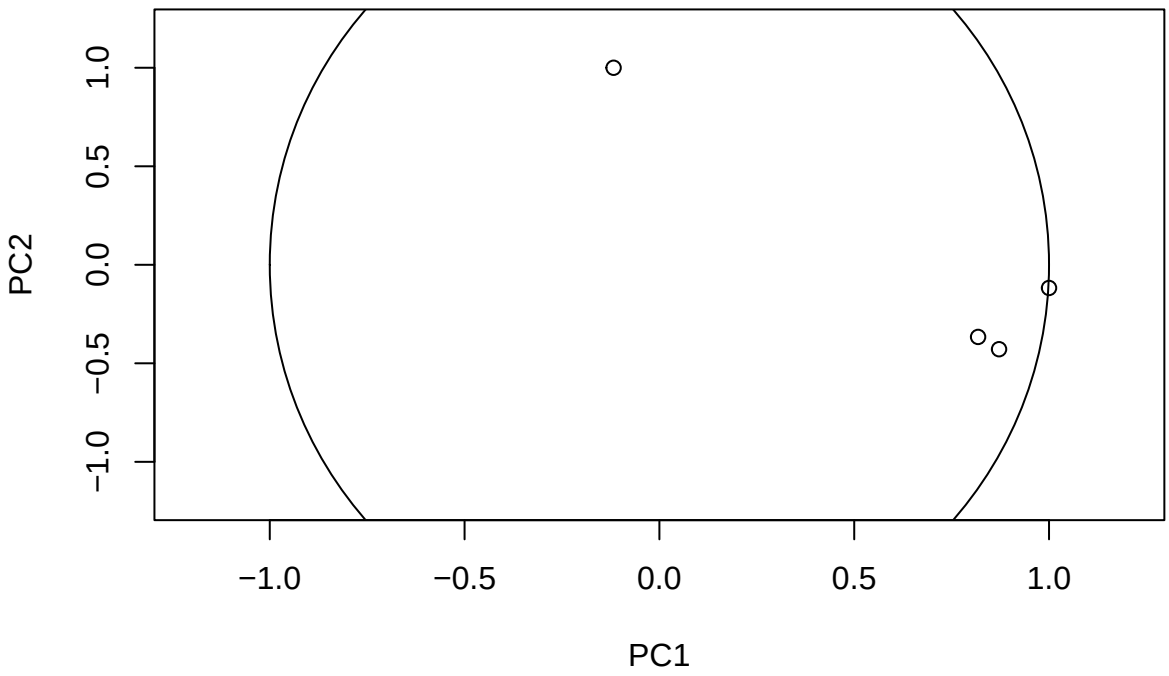
```
pca<-whiten(data,method="PCA")
head(pca)
```

```
##            L1       L2        L3          L4
## [1,] 1.370560 11.46196 -2.359181 -0.20137344
## [2,] 1.355962 10.45429 -3.011936  0.42539324
## [3,] 1.270930 10.51935 -2.195356 -0.08668731
## [4,] 1.340789 10.16746 -2.146514 -0.70555233
## [5,] 1.348874 11.47690 -1.937260 -0.61281589
## [6,] 1.566675 12.31848 -1.656210 -0.37278317
```

```
plot(pr$x[,1],pca[,1])
```




```
plot(cor(data),xlim=c(-1.2,1.2),ylim=c(-1.2,1.2),xlab="PC1",ylab="PC2")
symbols(0,0,circles=1,add=TRUE,inches=FALSE)
```


PC2
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
PC1